

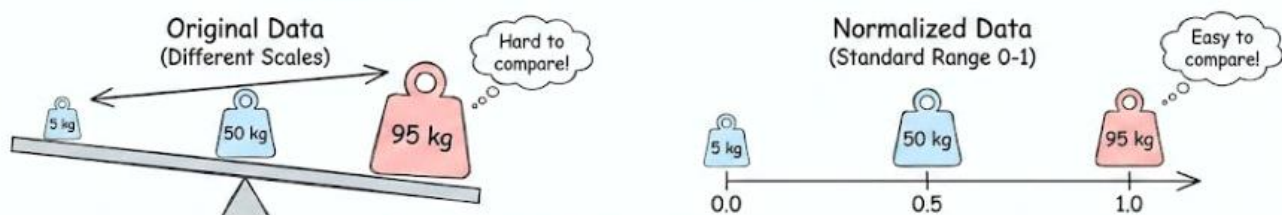
Normalisation

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Normalization is a technique often applied as part of data preparation for machine learning. The goal of normalization is to change the values of numeric columns in the dataset to use a common scale, without distorting differences in the ranges of values or losing information

Understanding Normalization (Min/Max Scaling)

The Goal: Resizing Data to a Standard Scale



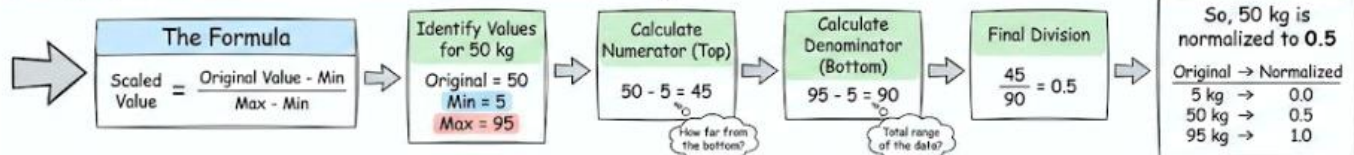
The Example: A Set of Weights

Our Data (Weights in kg): [5, 50, 95]

Minimum Value (Min) = 5

Maximum Value (Max) = 95

Step-by-Step Calculation (Let's normalize 50 kg)



Now all your weights are between 0 and 1, keeping their relative order! The smallest becomes 0, the largest becomes 1.

Normalization vs. Standardization

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When you have minimum and maximum value, then you can use min-max scaling.

Aspect	Normalization	Standardization
Objective	Rescale data to a fixed range (usually 0 to 1).	Rescale data to have a mean of 0 and a standard deviation of 1.
Formula	$X_{\text{norm}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$	$Z = (X - \mu) / \sigma$
Range of Transformed Data	[0, 1] or [-1, 1] (depending on the technique).	Data is centered around 0 with a standard deviation of 1.
Sensitivity to Outliers	Sensitive to outliers, as they affect the min and max values.	Less sensitive to outliers because the transformation uses mean and standard deviation.
Preferred Algorithms	Distance-based algorithms (e.g., k-NN, k-means, neural networks).	Algorithms that assume normally distributed data (e.g., linear regression, PCA, SVM).
When to Use	When features have known min and max values and when the model is distance-based.	When features have different scales, contain outliers, or the model assumes normally distributed data.